

Failure Mode & Condition Monitoring of a Telescopic Mobile Crane

Section 1 — Introduction

Mobile cranes operate under continuously changing conditions such as boom length, boom angle, working radius, hook load and outrigger configuration. Because the allowable lifting capacity changes with crane configuration and operating geometry, monitoring these parameters is important for identifying abnormal operating conditions.

The proposed Digital Twin represents the operating state of a telescopic mobile crane and uses simulated sensor parameters to evaluate its current condition. The system focuses on load, boom angle, boom length, working radius, outrigger condition and hydraulic pressure.

Section 2 — Selected Failure Mode

Failure Mode: Overload / Unsafe Operating Condition

An overload condition occurs when the current hook load exceeds the allowable rated capacity for the crane's current configuration and working radius.

The rated capacity of a mobile crane is not a single fixed value. It depends on parameters including boom length and working radius, with the crane's configuration also affecting the allowable load.

In the Digital Twin, the current operating parameters are evaluated continuously. The system calculates the working radius from boom geometry and compares the current hook load with the applicable rated capacity.

Section 3 — Causes

Cause	What happens
Excessive hook load	Load approaches/exceeds rated capacity
Increased working radius	Allowable lifting capacity decreases
Increased boom extension	Changes crane working geometry
Incorrect boom configuration	Crane operates outside intended chart condition
Improper outrigger deployment	Reduces support/stability
Incorrect operating condition	Actual condition may differ from assumed configuration

Section 4 — Sensors Used for Failure Detection

Parameter	Sensor
Hook/load	Load cell / load pin
Boom angle	Angle sensor / inclinometer
Boom length	Length sensor / encoder
Slewing angle	Rotary encoder
Hydraulic pressure	Pressure sensor
Crane inclination	Inclinometer / IMU
Outrigger position	Position/length sensor

Section 5 — How the Digital Twin Detects the Failure

CRANE OPERATING PARAMETERS

|

SENSOR / SIMULATION DATA

|

BOOM + LOAD + RADIUS

|

LOAD CHART LOOKUP

|

RATED CAPACITY

|

CURRENT LOAD / CAPACITY

|

CONDITION EVALUATION

|

NORMAL / WARNING / OVERLOAD

The Digital Twin receives the current crane state from the simulation parameters. Boom length and boom angle are used to determine the working geometry and working radius. The system then obtains the rated capacity from the selected load-chart data.

The current hook load is compared with the rated capacity. Load utilisation is calculated as:

$$\text{Load Utilisation} = \text{Current Hook Load} / \text{Rated Capacity} \times 100$$

The resulting value is displayed on the live telemetry panel so that the operator can observe how close the crane is to its rated operating limit.

Section 6 — Failure Detection Logic

Normal condition

Current load < rated capacity

→ Crane remains within the rated load condition.

Warning condition

Current load approaches rated capacity

→ Operator should review the crane configuration and load.

Overload condition

Current load > rated capacity

→ Digital Twin identifies an overload/unsafe operating condition.

Section 7 — Stability Failure

The deployed outriggers form a support polygon on the ground. The ground projection of the load position is evaluated relative to this polygon.

If the projected load position remains within the support polygon, the system considers the position geometrically supported. As the load approaches the polygon boundary, the system can provide a warning. If the load position moves outside the polygon, the system identifies a tipping condition.

Section 8 — Failure Mode → Digital Twin Response

Failure / abnormal condition	Parameter observed	Digital Twin response
Increasing load	Hook load	Load value updates
Increasing working radius	Boom length + angle	Radius changes
Approaching capacity	Load + rated capacity	Utilisation increases
Load exceeds rated capacity	Load / capacity	Overload condition
Poor outrigger configuration	Outrigger condition	Stability condition changes
Load moves toward support boundary	Slew angle + radius	Stability warning
Load moves outside support polygon	Load position	Tipping condition